RELIANCE INTERNATIONAL ACADEMY

Saraswatinagar, Kathmandu

SET I

Model set questions 2079

Grade: - XI | Full Marks: - 75

Subject: - Chemistry | Pass Marks: - 30

Stream: - Science | Time: - 3 hrs

Candidates are required to give answers in their own words as for as practicable. Figures in the margin indicate full marks. Do not scatter the answers.

Group – 'A'

Multiple choice questions

Attempt all questions. **[11¿ 1 = 11]**

1. Which one of the following compounds illustrate the law of multiple proportion?

a. H_2S and SO_2

b. $FeCl_2$ and $Fe(OH)_2$

c. NH_3 and NCl_3

d. CuO and Cu_2O

2. The angular momentum of electrons in nth orbit is given by

a. nh

b. h/2∏n

c. nh/2∏

d. $n^2h/2∏$

3. The elements were arrange in octave by

a. Dobereiner

b. Newland

c. Lother meyer

d. mandeleev

4. oxidation number of Fe in K4[Fe(CN)6] is

a. +2

b. -2

c. +4

d. +3

5. Nitrogen content of urea is

a. 26%

b. 36%

c. 46%

d. 64%

6. The boiling point is in the order

a. HF>HCl>HBr>HI

b. HF> HI>HBr>HCl

c. HCl> HBr> HI> HF

d. HCl> HF> HBr> HI

7. The metallurgical process in which the metal purified is

a. Roasting

b. Smelting

c. Electrolysis

d. floatation

8. which of the following does not produce hydrogen gas with caustic soda?

a. Tin

b. Silicon

c. Aluminium

d. White phosphorous

9. Which property is not common in organic compounds?

a. Low melting point

b. Slow chemical reaction

c. Isomerism

d. Fast chemical reaction

10. The IUPAC name of urea is

a. Diaminomethane

b. Carbamide

c. Aminomethanamide

d. carbogen

11. batch process is used for the manufacture of

a. petrol

b. cement

c. diesel

d. cosmetics

Group – 'B'
Short answer questions

Attempt All the questions **[5 x 8 = 40]**

1. what are quantum numbers? An atom has 20 electrons. Find out

i. it's atomic number and total number of p-electrons/

ii. the values of azimuthal quantum number(l) and magnetic quantum number(m) of 19th electron.

iii. It's group position in the periodic table. [1+1.5+1.5+1]

OR

Give a balanced chemical reaction for the preparation of chlorine in laboratory. How does Cl_2 reacts with conc. NaOH ? mention the uses of chlorine. [2 +1+2]

2. Define oxidation and reduction according to electronic concept giving example. Balance the following chemical equation by oxidation number or ion-electron method

$Zn + HNO_3 \longrightarrow Zn(NO_3)_2 + NO_2 + H_2O$ [2+3]

3. State and explain the law of mass action. Write the relation of Kc for the reaction

$N_2 + 3H_2 \longrightarrow 2NH_3$

OR

The kinetic theory of gases explains the macroscopic properties of gases, such as volume, pressure, and temperature as well as viscosity, thermal conductivity and mass diffusivity. The model also accounts for related phenomena such as Brownian motion.

i. Write the main postulates of kinetic theory of gas. [3]

ii. What is absolute zero temperature? [1]

iii. Calculate the number of moles of molecules present in 50 ml of an ideal gas exerting a pressure of 770 mm at $25^{o}C$? [1]

4. What are the necessary physical conditions and chemical principle involved in the manufacture of ammonia by Haber's process ? why is ammonia highly soluble in water ? [4 +1]
5. Depending upon the nature of minerals present in ores, calcinations and roasting are mainly used for the conversion of ores into their respective oxides.
   i. What do you mean by calcinations and roasting process in metallurgical process? [2]
   ii. Name the vessel in which the roasting is done [1]
   iii. Write the name of any two impurities that are removed in roasting. [2]
6. A simple organic compound may be divided into two parts alkyl group and functional group.
   i. Define functional group.
   ii. Write two characteristics of functional group.
   iii. Write the functional group of acid amide and carboxylic acid.
   iv. Write a homologous series containing –CHO as functional group with their IUPAC names. [1+1+1+2]
7. State the Huckel's Rules of aromaticity.A primary haloalkane (C_3H_7Br) undergoes wurtz reaction to give an alkane.
   i. Write the reaction involved.
   ii. What happens when P is heated with H_2 in presence of Ni?
   iii. What is soda lime decarboxylation? [1+1+1+2]
8. Sodium hydroxide is used in many industries: in the manufacture of pulp and paper, textiles, drinking water, soaps and detergents, and as drain cleaner. Historically, sodium hydroxide was produced by treating sodium carbonate with calcium hydroxide.
   i. Describe the manufacture of sodium hydroxide by diaphragm cell. (3)
   ii. Describe the cash flow in the production cycle of chemical industry. (2)

Group – 'C'
Long answer questions

Attempt all the questions **[3¿ 8 = 24]**

9. One volume of hydrogen reacts with one volume of Chlorine to give two volume of hydrogen Chloride gas. Which law of stoichiometry is illustrated? State the law. For a reaction, $Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + 2H_2O$. the reaction is carried out by mixing 7 gram of pure $Ca(OH)_2$ and 7 gram of pure NH_4Cl
   i. Find which one is limiting reactant and why?
   ii. Calculate the mole of unreacted reactant left over.
   iii. How many gram of $CaCl_2$ formed?
   iv. What volumes of NH_3 gas are produced at 27^oC and 1.5 atm pressure? [1+1+2+1+1+2]
   OR
   Define universal gas constant. Derive the ideal gas equation. A straight glass tube of length 25 cm has two inlets (A) and (B). NH_3 gas through the inlet (A)and HCl gas through the inlet (B) are allowed to enter the tube at the same time. White fumes of NH_4Cl appear at the point (M) inside the tube. Find the distance M from A. [1+3+4]
10. Write the principle and process with the self-explanatory diagram for the manufacture of sulphuric acid by contact process. What are the action of conc. sulphuric acid with (i) sucrose (ii) HI (iii) phosphorous. Give the test of sulphide ions in it's aqueous solution. [4+1+1+1+1]
11. Why it is necessary to prepare sodium extract for the detection of foreign elements in organic compounds? What is the composition of sodium extract if the organic compound contains nitrogen? How would you detect the presence of sulphur and halogen in organic compound? [2+2+4]

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SET II

Model set questions 2079

Grade: - XI | Full Marks: - 75

Subject: - Chemistry | Pass Marks: - 30

Stream: - Science | Time: - 3 hrs

Candidates are required to give answers in their own words as for as practicable. Figures in the margin indicate full marks. Do not scatter the answers.

Group – 'A'

Multiple choice questions

Attempt all the questions **[11¿ 1 = 11]**

1. Which of the following pairs of compounds illustrate the law of multiple proportions?

a. P_2O_3 and P_2O_5

b. P_2O_3 and B_2O_3

c. H_2S and H_2O

d. MO and Mo

2. Number of atoms present in 0.1 mole of water is

a. $0.1\ N_A$

b. $0.2N_A$

c. $0.3N_A$

d. $0.4N_A$

3. Which of the following statement is incorrect about the cathode rays

a. Cathode rays must be produced from the gas inside the discharge tube

b. The e/m value for cathode rays depends upon the nature of gas used and the nature of cathode

c. Cathode rays produced x-rays

d. Cathode rays affect photographic plates

4. Oxidation number of Fe in Fe3O4 is

a. +4/3

b. +2/3

c. +5/3

d. +8/3

5. laughing gas is

a. N_2O_3

b. N_2O

c. NO

d. N_2O_4

6. The acidic nature is in the order

a. HF> HCl> HBr>HI

b. HI> HBr> HCl> HF

c. HCl> HBr> HI> HF

d. HCl> HF> HBr> HI

7. Which of the following is a frother

a. Pine oil

b. NaCN

c. Xanthates

d. NaOH

8. Which of the following metal imparts golden yellow colour in Bunsen flame?

a. Natrium

b. Kalium

c. Lithium

d. Rubidium

9. Which property of carbon is responsible for the existence of large number and variety of organic compounds?

a. Covalent bonding

b. Tetrahedral nature

c. Catenation

d. Multiple bonding

10. The CHO group as prefix is

a. al

b. Formyl

c. Hydroxyl

d. carboxyl

11. Raw materials for the production of urea are

a. NH_3 and CO_2

b. N_2 and p

c. H_2 and P

d. NH_3 and p

Group – 'B'
Short answer questions

Attempt All the questions **[5 x 8 = 40]**

1. Rutherford's nuclear model of an atom is based on the alpha particles scattering experiment. During alpha particles scattering experiments, alpha

particles emitted from radioactive source is bombarded on thin gold foil and the fate of these particles were observed on movable zinc sulphide screen.

i. What are alpha particles?

ii. Which observation led Rutherford to conclude that (a) atom consist of positively charged mass called nucleus (b) most of the mass of atom is concentrated at the nucleus?

iii. What are the limitations of this model ? [1+2+2]

OR

Give the balanced chemical reaction for the preparation of HCl and HBr in the laboratory. How would you test the presence of Cl^- , Br^- and I^- ions in it's aqueous solution ? [2 +3]

2. Define oxidizing and reducing agents with examples. Balance the following chemical equation by oxidation number or ion electron method

$Zn + HNO_3 \longrightarrow Zn(NO_3)_2 + N_2O + H_2O$ [1+1+3]

3. Derive the relation between K_P and K_C.How does the temperature and pressure affect the state of equilibrium reaction,

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

OR

State and explain the Graham's law of diffusion. What are the reason behind the deviation of gases from their ideal behavior. What do you mean by real gas? [3+1 +1]

4. Give the balance chemical equation for the preparation of ammonia from nitrogen and hydrogen. What are the proper physical conditions for the better yield of ammonia? Write the action of ammonia on :

a. Mercurous nitrate paper

b. $CuSO_4$ solution

c. CuO [1+1+1+1+1]

5. Sodium is extracted by the electrolysis of a molten mixture of NaCl+ $CaCl_2$ in the ratio 2:3 using Down's cell.

i. Why is $CaCl_2$ mixed with NaCl for the extraction of Na?

ii. Name the materials used as cathode and anode in Down's cell. Also write the cathode and anode reactions.

iii. How is Na converted to NaOH? Write the reaction of aq. NaOH with ferric chloride solution? [1+2+2]

6. Describe the preparation of ethane in laboratory. What products would you expect when 2-bromopropane is heated with

i. Na/dry ether

ii. With Alc.KOH [3+1+1]

7. What is meant by homologous series and write a series containing alcohol as a functional group. Write the structural formula and IUPAC name of:

i. 4th member of carboxylic acid

ii. 2nd member of amide

iii. 3rd member of acid chloride

iv. 2nd member of alcohol [1+2+1+1]

8. What is Haber's process? Explain it's importance in applied chemistry. Write down the differences between batch production system and continuous production system. [1 +2+2]

Group – 'C'
Long answer questions

Attempt all the questions **[3¿ 8 = 24]**

9. State the law of multiple proportion. An oxide of nitrogen contains it's half volume of nitrogen and it's vapour density is 15. Determine it's molecular formula. 5 gram of pure $CaCO_3$ is treated with 5 gram of 5 gram of HCl to produce $CaCl_2$, H_2O and CO_2:

I. Find which one is limiting reactant.

II. Calculate the mass of $CaCl_2$ formed

III. How many number of water molecules are produced?

IV. Calculate the volume of CO_2 produced at NTP? [1+2+1+1+1+2]

OR

What do you mean by ideal gas? Mention the proper conditions under which the real gas behaves as an ideal gas? State and explain the pressure volume relationship of gas law. How long will it take 500 ml of hydrogen gas to diffuse through a partition if 250 ml. of oxygen diffuse in 50 minutes under similar conditions? [1+1+3+3]

10. Describe the principle and self explanatory sketch for the manufacture of ammonia by Haber's process. How does ammonia react with:

i. Carbon dioxide
ii. Copper sulphate solution
iii. Sodium metal
iv. Chlorine gas in excess. [4+1+1+1+1]

11. For the detection of elements N, halogens, S in organic compounds, why is sodium fusion carried out? How would you detect the presence of nitrogen in organic compounds? Also explain, why the sodium extract is alkaline in nature? [2+4+2]

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